

TrES-1b

R-2187

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_110302 · created 2026-08-13T23:27:24+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455622.98248 +/- 0.00065 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1353 +/- 0.0084
Transit depth (Rp/Rs)^2	1.83 +/- 0.23 [%]
Orbital Inclination (inc)	89.15 +/- 0.7
Airmass coefficient 1 (a1)	1.3553 +/- 0.0084
Airmass coefficient 2 (a2)	-0.1131 +/- 0.0022
Scatter in the residuals of the lightcurve fit is	0.61 %
Best Comparison Star	None
Optimal Aperture	4.08
Optimal Annulus	12.71
Transit Duration (day)	0.1027 +/- 0.0021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1353 ± 0.0084 vs 0.1358 (deviation 0.05σ)
Duration fit vs published	0.1027 d vs 0.1042 d (deviation 0.61σ)
Mid-time O–C	-1.7 min
Depth SNR	14.0
Residual scatter	0.61 %

Light curve

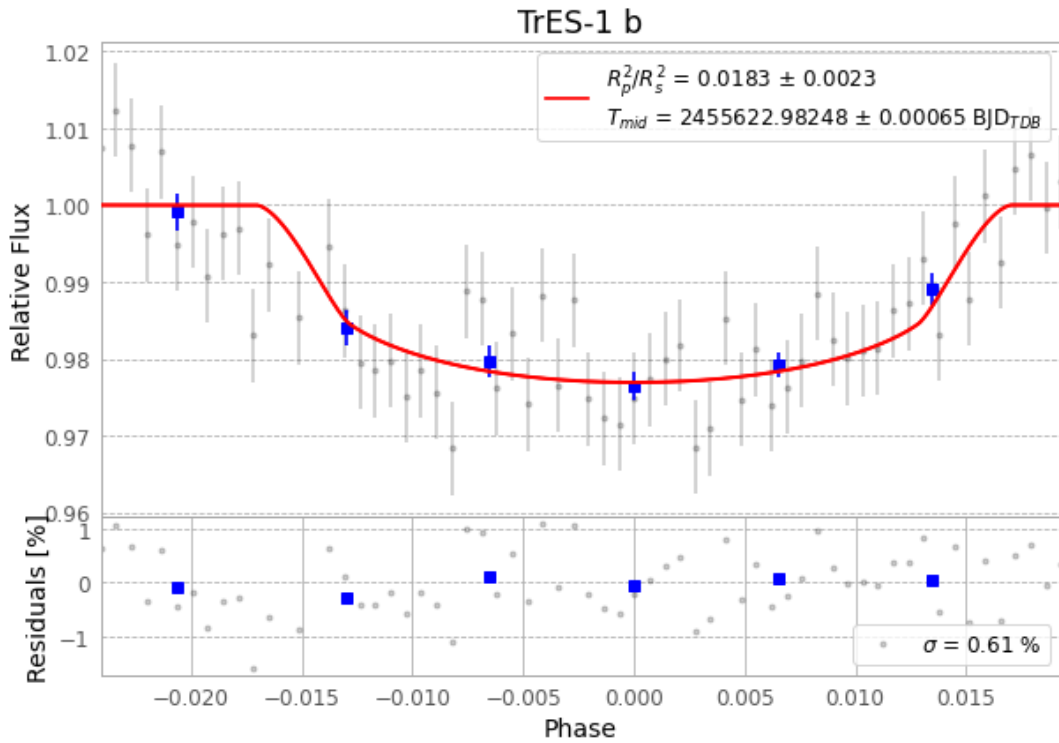


Figure 1 — FinalLightCurve_TrES-1 b_2011-03-02.png (SHA-256 0a8cea3a0c1810a3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [264, 140], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0d86ddade656926e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

64 FITS frames (42 MB), TRES-1110302095113.FITS → TRES-1110302130013.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-110302.log (4ee492c4855b...), FinalLightCurve_TrES-1 b_2011-03-02.png (0a8cea3a0c18...), FinalLightCurve_TrES-1 b_2011-03-02.csv (4f7e2b345b46...)

Compute resources

Wall time 112.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-13 23:27Z.